WORKED SOLUTION

STEP II 2012, Question 1

Pure • Binomial Series Coefficients

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q1

The key idea

Each factor $(1-x^k)^{-1}$ is a geometric series in x^k, so multiplying such factors together and reading off the coefficient of x^n amounts to *counting* the ways of writing n as a sum of the allowed pieces. Once you see the coefficients as counts, the case analysis organises itself, and multiplying by $(1-x)^{-1}$ simply turns a list of coefficients into its running totals.

Solution

The general term

By the binomial expansion, $(1-y)^{-2} = \sum_{r\geq 0}(r+1)y^r$ for $|y|<1$ (the coefficient is $\binom{-2}{r}(-1)^r = r+1$). Putting $y = x^6$,

$$(1-x^6)^{-2} = \sum_{r=0}^{\infty}(r+1)\,x^{6r},$$

so the general term is $(r+1)x^{6r}$, and every power of x that is not a multiple of 6 has coefficient 0.

Part (i)

Also $(1-x^3)^{-1} = \sum_{s\geq 0} x^{3s}$. Multiplying the two series, a term x^n arises from each pair (r,s) of non-negative integers with $6r+3s=n$, contributing $(r+1)\cdot 1$. So

$$\text{coefficient of } x^n = \sum_{\substack{r,s\geq 0\\ 6r+3s=n}} (r+1).$$

For $n=24$ we need $2r+s=8$, so r can be $0,1,2,3,4$ (each with exactly one $s=8-2r\geq 0$). The coefficient of x^{24} is

$$1+2+3+4+5 = 15.$$

General n. Since $6r+3s$ is always a multiple of 3, the coefficient is 0 unless $3 \mid n$. If $n = 3m$, we need $2r + s = m$, and for each r with $0 \le 2r \le m$ there is exactly one valid s. So r runs from 0 to $\lfloor m/2 \rfloor$, and the coefficient is

$$1 + 2 + \cdots + (\lfloor m/2 \rfloor + 1).$$

Writing n according to its remainder on division by 6:

- $n = 6k$ (so $m = 2k$, $\lfloor m/2 \rfloor = k$): coefficient $\frac{1}{2}(k+1)(k+2)$;
- $n = 6k+3$ (so $m = 2k+1$, $\lfloor m/2 \rfloor = k$): coefficient $\frac{1}{2}(k+1)(k+2)$ as well;
- $n \not\equiv 0 \pmod 3$: coefficient 0.

As a check, $n = 24 = 6 \times 4$ gives $\frac{1}{2} \cdot 5 \cdot 6 = 15$, as found above.

Part (ii)

Write b_n for the coefficient of x^n found in part (i), and c_n for the coefficient we now want. Since $(1-x)^{-1} = 1 + x + x^2 + \cdots$,

$$\Big(\sum_j b_j x^j\Big)(1 + x + x^2 + \cdots) = \sum_n \Big(\sum_{j=0}^{n} b_j\Big) x^n,$$

so $c_n = b_0 + b_1 + \cdots + b_n$: the running total of the part (i) coefficients.

Ask yourself Why does multiplying by $(1-x)^{-1}$ give running totals?

To get x^n in the product we take x^j from the first series and x^{n-j} from $1 + x + x^2 + \cdots$, whose coefficient is always 1. So every b_j with $0 \le j \le n$ appears exactly once in c_n, with weight 1.

Only b_j with $3 \mid j$ are non-zero, and by part (i) they come in equal pairs: $b_{6k} = b_{6k+3} = T_{k+1}$, where $T_p = \frac{1}{2}p(p+1)$ is the pth triangular number. So for $n = 24$,

$$\begin{aligned} c_{24} &= (b_0 + b_3) + (b_6 + b_9) + (b_{12} + b_{15}) + (b_{18} + b_{21}) + b_{24} \\ &= 2(1 + 3 + 6 + 10) + 15 = 40 + 15 = 55, \end{aligned}$$

as required.

For $n = 25$, the extra term b_{25} is 0 because $3 \nmid 25$, so $c_{25} = c_{24} = 55$.

For $n = 66 = 6 \times 11$, the same pattern gives

$$c_{66} = 2\,(T_1 + T_2 + \cdots + T_{11}) + T_{12}.$$

The sum of the first p triangular numbers is the tetrahedral number $\frac{1}{6}p(p+1)(p+2)$ (by induction, or because $T_1 + \cdots + T_p = \binom{p+2}{3}$ by the hockey-stick identity). So $T_1 + \cdots + T_{11} = \frac{1}{6} \cdot 11 \cdot 12 \cdot 13 = 286$, and $T_{12} = 78$. Hence

$$c_{66} = 2 \times 286 + 78 = 650.$$

The same argument gives a general formula: if $n = 6K + r$ with $0 \leq r \leq 2$ then $c_n = 2\binom{K+2}{3} + \binom{K+2}{2}$, and if $3 \leq r \leq 5$ then $c_n = 2\binom{K+3}{3}$. With $K = 4$ the first gives $40 + 15 = 55$; with $K = 11$ it gives $572 + 78 = 650$.

Checks

The coefficient of x^n in part (ii) counts the non-negative solutions of $6a + 6b + 3c + d = n$ (one geometric series for each factor, with $(1 - x^6)^{-2}$ counted as two). A direct computer count of these solutions gives 55, 55 and 650 for $n = 24, 25, 66$. Multiplying out the series term by term up to x^{80} also confirms the formulas for both parts. By hand, small cases are quick: in part (ii), $c_0 = c_1 = c_2 = 1$ and $c_3 = c_4 = c_5 = 2$, matching the general formula with $K = 0$.

Common slips

- Forgetting the case $n = 6k + 3$ in part (i): odd multiples of 3 do appear, from an odd number of factors of x^3.
- Writing the general term of $(1 - x^6)^{-2}$ as $r\,x^{6r}$ instead of $(r + 1)x^{6r}$; the constant term is 1, not 0.
- Treating x^{25} as a fresh calculation: it equals the x^{24} coefficient, because no term of the part (i) series has exponent 25.

Beyond STEP

Power series used as counting devices like this are called *generating functions*; the product here counts the ways to pay n pence with coins worth 6, 6 (of a different design), 3 and 1. The theory of partitions grows from exactly this idea, and the fact that the answers are piecewise polynomials, depending on n modulo 6, is a general phenomenon ("quasi-polynomials"). For a lively introduction to the method, read Wilf, *generatingfunctionology*.

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